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Steven Kelly

YPFS, Yale School of Management

Vincient Arnold

YPFS, Yale School of Management

Greg Feldberg

YPFS, Yale School of Management

Andrew Metrick

YPFS, Yale School of Management

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Ad Hoc Emergency Liquidity Programs in the 21st Century¹

Steven Kelly,² Vincient Arnold,³ Greg Feldberg,⁴ and Andrew Metrick^{5, 6}

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Abstract

This paper surveys 22 case studies of 21st century instances when financial crisis-fighters implemented ad hoc emergency liquidity (AHEL) interventions, interventions designed to provide liquidity to a troubled institution that the authorities believe is systemically important. While emergency liquidity support is often introduced with the real or communicated intention of preventing illiquidity from leading to insolvency, the liquidity crisis should instead be viewed as the manifestation of the market's assessing the firm as nonviable as a going concern. For that reason, authorities should provide AHEL assistance only to institutions that they have deemed viable or that they have committed to make viable through additional interventions, most commonly through a government capital injection or merger with a stronger institution. Despite AHEL assistance, which the authorities in several cases sized to meet all potential funding outflows from the troubled firms, in no cases did liquidity provision alone prove a "cure" to the run on the institution. Crisis-fighters also should not use the terms of an AHEL intervention to manage moral hazard. Moral hazard can be addressed in the more structural policy responses that will need to follow AHEL assistance. AHEL programs should focus on providing sufficient liquidity to get the institution through its acute crisis phase.

Keywords: ad hoc emergency liquidity, Bagehot's dictum, crisis-fighting, financial stability, lender of last resort, solvency, viability

¹ This survey reviews a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering ad hoc emergency liquidity provision. The individual cases underlying it are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/>. Wiggins, Fulmer et al. (2023) surveys broad-based emergency liquidity programs.

² Associate Director of Research, YPFS, Yale School of Management.

³ Research Associate, YPFS, Yale School of Management.

⁴ Research Scholar and Director of Research, YPFS, Yale School of Management.

⁵ Janet L. Yellen Professor of Finance and Management and YPFS Program Director, Yale School of Management.

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Introductory Note: This survey is an analysis of key considerations for policymakers seeking to establish an ad hoc emergency liquidity (AHEL) program—that is, emergency liquidity targeted at an individual institution. It is based on insights derived from 22 Yale Program on Financial Stability case studies on modern AHEL interventions and from the existing literature on the topic. The cases were chosen from the Metrick-Schmelzing Banking Crisis Intervention Database,⁷ from which we selected all documented 21st century AHEL programs with sufficient information available to write a case study and that were not (initially) intended as resolution financing; see Figure 1 for the full list of underlying cases. While this survey can help inform a decision about whether or not to establish an AHEL program, our main purpose is to assist policymakers who have already made that decision in designing the most effective program possible. In analyzing the programs that are the focus of this survey, we used a color-coded system to highlight certain particularly noteworthy design features.

Treatment	Meaning
BLUE – INTERESTING	A design feature that is interesting and that policymakers may want to consider. Typically, this determination is based on the observation that the design feature involves a unique and potentially promising way of addressing a challenge common to this type of program that may not be obvious. Less commonly, empirical evidence or a consensus will indicate that the design feature was effective in this context, in which case we will describe that evidence or consensus.
YELLOW – CAUTION INDICATED	A design feature that policymakers should exercise caution in considering. Typically, this determination is based on the observation that the designers of the feature later made significant changes to the feature with the intention of improving the program. Less commonly, empirical evidence or a consensus will indicate that the design feature was ineffective in this context, in which case we will describe that evidence or consensus.

This highlighting is not intended to be dispositive. The fact that a design feature is not highlighted or is highlighted blue does not mean that it should always be considered or will be effective under all circumstances. Similarly, the fact that a design feature is not highlighted or is highlighted yellow does not mean that it should not be considered or that it will never be effective under any circumstances. The highlighting is our subjective attempt to guide readers toward certain design features that (1) may not be obvious but are worth considering or (2) require caution.

⁷ Metrick and Schmelzing (2024).

Introduction

This survey analyzes 12 Key Design Decisions of 22 underlying case studies of modern-day emergency liquidity interventions targeted at specific institutions, which we call ad hoc emergency liquidity (AHEL) programs. See Figure 1 for the 22 interventions surveyed here.⁸

When faced with sudden market demands for liquidity, central banks typically prefer to provide it through open market operations or broadly available programs. However, such programs are not always sufficient to provide the desired amount of liquidity to individual financial institutions suffering from a viability crisis, so an AHEL program becomes necessary. Similar to what might be commonly referred to as a solvency crisis, we refer to a viability crisis as a situation in which creditors or broader financial markets have lost confidence in an institution's ability to continue as a going concern and either do not expect authorities to intervene to save it or do not expect such interventions to restore the institution's going-concern status.

When a failing institution has systemic implications and other liquidity facilities are unavailable to provide additional liquidity, policymakers may provide assistance through AHEL programs. Academic literature, policy memos, and official speeches have broadly focused on “emergency liquidity assistance” programs, tending to address ad hoc and broad-based emergency liquidity (BBEL) collectively.⁹ We have analyzed them separately. This survey seeks to supplement those materials with an in-depth treatment of the less-acknowledged, but common, case of needing to design an AHEL assistance program. For those interested in designing broader programs, see the sister Yale Program on Financial Stability surveys on BBEL interventions (Wiggins, Fulmer, et al. 2023) and market-support liquidity interventions (Rhee, Feldberg, et al. 2020; Rhee, Engbith, et al. 2022).

⁸ Twenty-two institutions received AHEL assistance in our study. AIG is the subject of two of our underlying cases—AIG RCF and AIG SLF—because it received two discrete AHEL lines; however, it is counted only once in our list of 22 AHEL programs. HBOS and RBS are counted separately because they are separate institutions, but they are grouped together in one underlying case study—HBOS and RBS—because the respective AHEL programs were substantively similar.

⁹ A welcome exception is a 2016 International Monetary Fund (IMF) paper that, similar to our focus on AHEL programs, centered on what its authors call “idiosyncratic support” (Dobler et al. 2016).

Figure 1: Case Studies and Short-form Case Names

Case Name	Short-form Case Name	Reference
Brazil: Banco BTG Pactual Emergency Liquidity Program, 2015	Brazil–BTG Pactual	(Arnold 2025b)
Cyprus: Laiki Bank Emergency Liquidity Assistance, 2011	Cyprus–Laiki	(Schaefer-Brown 2025)
Denmark: Roskilde Bank Emergency Liquidity Program, 2008	Denmark–Roskilde	(Decker 2025a)
Germany: IKB Deutsche Industriebank Emergency Liquidity Program, 2008	Germany–IKB	(Alden and George 2025)
Iceland: Kaupthing Emergency Liquidity Program, 2008	Iceland–Kaupthing	(Alden and Brougher 2025)
Indonesia: Bank Century Emergency Liquidity Program, 2008	Indonesia–Bank Century	(Arnold 2025c)
Ireland: Anglo Irish Bank Emergency Liquidity Assistance, 2009	Ireland–Anglo Irish	(Gupta, Khairallah, and Nik Mokhtar 2025)
Latvia: Parex Bank Emergency Liquidity Program, 2008	Latvia–Parex	(Decker 2025b)
Moldova: Consortium of Banks Emergency Liquidity Program, 2014	Moldova–Consortium	(Arnold 2025d)
Russia: Otkritie Emergency Liquidity Program, 2017	Russia–Otkritie	(Hoffner 2025)
Spain: Caja de Ahorros Castilla–La Mancha Emergency Liquidity Assistance, 2009	Spain–CCM	(Arnold and Swaminathan 2025)
Sweden: Carnegie Investment Bank AB Emergency Liquidity Program, 2008	Sweden–Carnegie	(Alden and French 2025)
Switzerland: Credit Suisse Emergency Liquidity Program, 2023	Switzerland–Credit Suisse	(French, Heaphy, and Kelly 2025)
United Kingdom: HBOS and RBS Emergency Liquidity Program, 2008	UK–HBOS and RBS	(Decker and French 2025)
United Kingdom: Northern Rock Emergency Liquidity Program, 2007	UK–Northern Rock	(Decker, French, and Shyu 2025)
United States: The Rescue of American International Group Module A: The Revolving Credit Facility	US–AIG RCF	(Buchholtz and Lawson 2021)
United States: The Rescue of American International Group Module B: The Securities Borrowing Facility	US–AIG SBF	(Engbith, Buchholtz, and Jeffereis 2021)
United States: Bank of America Emergency Liquidity Program, 2009	US–Bank of America	(Arnold 2025e)
United States: Bear Stearns Emergency Liquidity Program, 2008	US–Bear Stearns	(Arnold 2025f)
United States: Citigroup Emergency Liquidity Program, 2008	US–Citigroup	(Arnold 2025g)
United States: First Republic Bank Emergency Liquidity Program, 2023	US–First Republic	(Gupta, French, and Kelly 2025)
United States: Lehman Brothers Broker-Dealer Emergency Liquidity Program, 2008	US–Lehman LBI	(George and Kelly 2025)

Source: Authors' compilation.

Crisis-fighters typically resort to an AHEL program quickly when addressing the acute, or “panic,” phase of a crisis at a troubled financial institution, as an AHEL intervention is generally the quickest possible policy response to implement and can help the institution meet the demands of a run. However, even an optimally designed and executed AHEL

program should not be expected to prevent or solve the chronic, or “debt overhang,” phase of the institution’s crisis. Addressing an institution’s chronic problems typically demands more structural policy responses—such as capital injections, liability guarantees, balance sheet restructurings, or orderly resolutions. Crisis-fighters often suggest that their intent in providing emergency liquidity is to prevent an institution that is “illiquid but solvent” from tipping into insolvency, but liquidity crises are rare in the absence of deeper-seated solvency or viability concerns.

In other words, illiquidity in the case of an individual institution is almost always the manifestation of the market’s assessment of the institution’s nonviability as a going concern.¹⁰ An assessment of nonviability may simply reflect a view that the bank is insolvent based on depressed market valuations, rather than book valuations. In other cases, it may reflect a view that a bank, solvent or not, no longer has a sustainable business model—perhaps due to macroeconomic developments; competitive dynamics; regulatory problems; or the loss of key customers, staff, or important licenses or charters (BIS CGFS 2017; Dobler et al. 2016; Kelly and Rose 2025).

Indeed, in none of the surveyed cases did the illiquidity simply pass after the institution received liquidity assistance and leave the AHEL borrower’s balance sheet relatively unchanged in its wake. All cases required significant balance sheet restructurings, almost always with the assistance of additional policy interventions. The experiences of these cases yield or support several different insights, which we discuss further in the sections that follow. Crisis-fighters should design ad hoc emergency liquidity provision to buy time to address fundamental viability issues at the borrower. They shouldn’t expect these programs to help a bank avoid insolvency by addressing illiquidity. Other policy interventions will almost certainly be necessary and should be rolled out as expeditiously as possible following the AHEL intervention. Recognizing that AHEL assistance can meet the problems of the institution’s acute crisis phase only, rather than solve its chronic issues, policymakers can spend less of their focus on managing moral hazard when designing AHEL programs. Moral hazard can be addressed in the more structural policy responses that will need to follow; AHEL programs should focus on providing sufficient liquidity, particularly if the institution has been deemed viable (inclusive of the impact of other forthcoming interventions).

Liquidity versus Solvency—versus Viability

Modern crisis-fighters regularly return to Walter Bagehot’s famous 1873 dictum on the appropriate role for a central bank during a crisis, which is typically stated pithily as “lend freely, against good collateral, at a penalty rate.” Many authors have also either ascribed a “solvency” requirement to Bagehot¹¹ or added it themselves, updating the dictum to something along the lines of “lend freely to solvent counterparties, against good collateral,

¹⁰ For related recent literature, see Baron, Verner, and Xiong (2021) and Correia, Luck, and Verner (2024).

¹¹ Indeed, Bagehot did not believe in bailing out insolvent firms—he had no argument with the Bank of England’s refusal to lend to the insolvent Overend, Gurney & Company, for example. He fully supported the Bank of England’s broad-based extensions of credit to many market participants to stem the contagion following Overend Gurney’s failure, however (Bagehot 1873).

at a penalty rate.”¹² It has thus long been a common view among financial authorities that central banks should provide liquidity only to firms that are “solvent but illiquid.”

Yet, when possible, central banks in the case studies we surveyed often did not constrain themselves to thresholds of literal, or accounting, solvency when confronted with the potential for systemic failure. Rather, they tended to evaluate more broadly the systemically important financial institution’s viability as a going concern, inclusive of the effect of the additional intended policy interventions. However, even in cases where central banks have shifted their financial assessment of potential borrowers to focus on their viability, the relevant regulations and statutes tend to still focus on “solvency” (BIS CGFS 2017; Dobler et al. 2016).

During a crisis, market participants may assess an accounting-insolvent firm to be viable as a going concern if they trust its management’s business model or they trust the authorities to provide sufficient support. They may also assess an accounting-solvent firm to be nonviable in the absence of those things.¹³

From the authorities’ perspective, they need to consider whether their resources and available tools will be sufficient to restore a firm’s viability as a going concern. As a Bank for International Settlements (BIS) working group writes in a 2017 paper,

While there is no formal definition of viability, the concept requires the assessor to look beyond the current valuation of the firm’s assets and liabilities and to consider the firm’s and/or relevant authorities’ ability to reestablish the firm as a going concern, possibly with the help of [liquidity assistance] and after adjustments in its business model. (BIS CGFS 2017)

Practically speaking, balance sheet solvency is particularly difficult to evaluate in real time when considering the speed of a crisis, the delays and uncertainties of financial accounting, and the endogeneity of asset and franchise values to crisis responses (Dobler et al. 2016; Goodhart 1999; Kelly 2024a). Moreover, even if the central bank is able to ascertain solvency, but the firm is nonviable, the central bank may realize the risks typically ascribed to lending to insolvent institutions. That is, lending to nonviable firms increases the likelihood, relative to lending to an insolvent-but-viable firm, of the central bank’s facing policy drawbacks such as: experiencing losses; reallocating losses between stakeholders; needing to take possession of too much, or difficult-to-manage, collateral; adding stigma to other firms taking liquidity assistance; or effectively drifting into fiscal policy or other political decision-making (Geithner 2019; Tucker 2014; Tucker 2020). By contrast, focusing on a systemically important firm’s long-term viability, assuming whatever government support is financially

¹² See, for example, Bernanke (2015b), Jones (2023), Madigan (2009), Praet (2016), and Tucker (2014). For a much earlier example, see Schwartz (1992), which refers to the “ancient injunction to central banks to lend only to illiquid banks, not to insolvent ones.”

¹³ As Dobler et al. (2016) notes, “A viability assessment is ‘business model’-focused, undertaken to determine that the entity can reasonably be expected to have continued potential for generating sufficient cash flow to repay the” central bank. It adds, “A bank that has sufficient capital today but is expected to run losses for the foreseeable future would not be viable.”

and politically feasible, can avoid those costs by resulting in loans only to firms that are truly going concerns.

When an ad hoc emergency liquidity program is required for a troubled, systemically important firm, the source is almost never some contagion of illiquidity only, but rather the market's fears about the firm's viability as a going concern. Policymakers should thus take care to not misdiagnose the run on an institution's liquidity as the cause rather than the effect of its problems—which would also cause an assessment that backstop liquidity is all that is required.

Indeed, several of the AHEL interventions surveyed here demonstrate instances where crisis-fighters sized the AHEL assistance to effectively cover all possible short-term outflows, yet counterparty and customer outflows continued until additional measures were implemented. Clearly, even assuring repayment of all maturing obligations is not a sufficient condition to stop runs.¹⁴ Where we do find success for AHEL interventions is in buying time to implement the more structural responses necessary to address an institution's viability concerns—such as capital injections, liability guarantees, balance sheet restructurings, and the like—which often require relatively more planning or time-consuming measures.

Moral Hazard in Ad Hoc Liquidity Provision

Given this context, it is important to reconsider whether all policy trade-offs associated with financial institution rescues need to be carefully weighed during the design of the AHEL program specifically; often, rescue terms intended to limit moral hazard are best left to the policy interventions that inevitably follow. The case studies examined here frequently show that when AHEL program terms are initially established as overly punitive because of moral hazard concerns, they often later need to be loosened for the intervention to be a sufficient and effective bridge to the more structural policy responses that follow. Moreover, in the case of a borrower with access to a standing liquidity facility, the very presence of an AHEL program is often indicative of insufficiently generous terms at the standing facility to provide enough liquidity to the borrower. The AHEL intervention must, by design, be relatively more expansive than a standing facility.

As such, design decisions for an AHEL program should depart from best practices for standing facilities and BBEL programs. Returning to Bagehot's dictum, which advises that when broad-based liquidity provision is needed, central banks should "lend freely, against good collateral, at a penalty rate," it seems only "lend freely" survives in practice when designing effective AHEL interventions.

At a penalty rate? Although applying a penalty rate to emergency lending is often suggested to discourage moral hazard and encourage repayment as financial conditions normalize, doing so may be counterproductive when an AHEL intervention is necessary—that is, when the market has targeted a run on a particular institution. A penalty rate in such a case is liable to instead accelerate the drain on the affected institution's financial resources (and be

¹⁴ See also Kelly and Rose (2025).

disfavored by the market and the rating agencies)—thus intensifying crisis-fighters’ race against the clock. Given that AHEL interventions effectively work only as bridges to more structural policy responses, it behooves crisis-fighters to save the more punitive or costly design features for those later-stage responses, when the institution has been stabilized as a going concern.

Against good collateral? Our case studies consistently demonstrate that, in the AHEL instance, collateral sufficiency tends to be interpreted as flexibly as is necessary for the central bank (or other lender¹⁵) to get sufficient liquidity to the foundering institution. While it was typically only when the prospective borrower was unable to post sufficient collateral that policymakers deviated from the standard of requiring sufficient assets to (over)collateralize the loan, this departure should not be interpreted as the ubiquitous disregard of caution by desperate crisis-fighters. Rather, if the AHEL intervention is known to be a bridge to other policy responses, those policies themselves can provide additional security. For instance, if the AHEL-receiving institution is going to merge with a healthy institution, the AHEL lender can consider the strength of the acquirer’s balance sheet. Or, if the government is organizing a capital injection, the AHEL lender can consider such incoming funds when evaluating the borrower’s financial condition and ability to repay. The risk of such designs, to be sure, is in the level of uncertainty of the forthcoming additional policy responses.

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The rest of this article surveys the 22 AHEL intervention case studies across 12 Key Design Decisions, codifying best practices for each design component. Throughout, we highlight specific examples of design features that were particularly effective (in blue) and those that should warrant additional caution (yellow).

¹⁵ As discussed in Key Design Decision No. 7, Source and Size of Funding, AHEL programs are sometimes, though infrequently, effected by non-central bank lenders. Throughout the survey, we use the terms “AHEL lender” and “central bank” interchangeably; yet, if a detail discussed is specific to central banks and doesn’t apply to non-central bank lenders, we say so clearly.

Key Design Decisions

1. Purpose: Why did authorities choose to implement an ad hoc emergency liquidity intervention?

Systemic Importance

In almost all cases, the driving motivation for extending AHEL assistance was to ameliorate funding pressures at a systemically important institution, as a bridge to more permanent policy solutions (see Key Design Decision No. 2, Part of Package). An institution's perceived systemic importance was always a necessary condition for receiving AHEL assistance. While size relative to the financial system was a common metric for systemic importance determinations, authorities frequently invoked characteristics other than size alone—such as interconnectedness or other factors that could cause contagion. Thus, the institution may not have been generally considered systemic *ex ante* but was instead deemed systemic at the time of needing the assistance, owing to the surrounding acute crisis and threat of greater systemic fallout in the absence of AHEL assistance.¹⁶

In the case of Bear Stearns, then-President of the Federal Reserve Bank of New York Timothy Geithner said that “Bear was not that big—only the seventeenth largest U.S. financial institution at the time—but it was completely enmeshed in the fabric of the system,” and that it was less a case of “too big to fail” than “too interconnected to fail” (Geithner 2014, 150–51). In other cases, authorities pointed to market signals as an indication of the institution's systemic importance. In providing AHEL assistance to the relatively small Bank Century in 2008, Indonesian authorities said that financial markets were so unstable at the time that the failure of Bank Century would have triggered a systemwide bank run. In the Banking Crisis of 2023, a consortium of banks provided AHEL assistance to midsized First Republic to publicly express confidence in the US financial system and prevent a First Republic failure from taking down additional banks.

Limitations of Standing or Other Broad-Based Facilities

In most jurisdictions, standing discount window, or “Lombard,” facilities are the first line of defense for a central bank to address sudden spikes in liquidity demand. When these prove inadequate, a central bank will typically open new, broad-based emergency liquidity facilities to expand systemwide liquidity (Wiggins, Fulmer, et al. 2023).¹⁷ A primary driver for policymakers to establish an AHEL program was the borrower's inability to obtain the necessary liquidity through standing or BBEL facilities. Systemically important nonbanks would need AHEL programs simply because standing facilities are usually available only to banks. Banks and other institutions eligible for standing or BBEL facilities, though, sometimes need AHEL programs because they lack sufficient eligible collateral, the requisite

¹⁶ This idea is conceptually similar to what the Financial Stability Board has, in a resolution context, dubbed “banks systemic in failure” (FSB 2024).

¹⁷ They may also provide support through similar programs to backstop other funding markets such as those for commercial paper or repurchase agreements (repos); see Rhee, Feldberg, et al. (2020) and Rhee, Engbith, et al. (2022).

financial health, or adequate borrowing preparations to obtain sufficient liquidity from those broad-based facilities. In our cases, AHEL facilities were thus frequently supplements to standing and BBEL facilities once the borrower had used the latter facilities to the fullest extent possible.

In the Eurosystem, for example, the European Central Bank (ECB) maintains a list of acceptable collateral, but national central banks (NCBs) can accept a broader range of collateral when providing AHEL assistance—which is implemented under a dedicated emergency liquidity assistance (ELA) protocol that is separate from the Eurosystem’s general emergency liquidity provision. In most of the surveyed cases of Eurosystem ELA, lacking sufficient ECB-eligible collateral was the primary factor pushing the banks to obtain NCB ELA (Cyprus–Laiki, Ireland–Anglo Irish, and Spain–CCM). This dynamic played out in numerous other countries, such as the UK (where HBOS and RBS had exhausted collateral eligible for BBEL facilities) and Switzerland (where Credit Suisse [CS] had exhausted its access to both major BBEL facilities).

Bank Indonesia’s experience with Bank Century illustrates that collateral policies were not the only policies limiting access to BBEL facilities. In that case, regulatory capital requirements disqualified Bank Century from standing facilities and drove Bank Indonesia’s bespoke arrangement for Bank Century.¹⁸

2. Part of a Package: What other measures typically accompanied AHEL interventions?

For none of the 22 AHEL assistance recipients was the AHEL support a stand-alone solution—that is, it was never just a solution to a temporary liquidity shortage that later righted itself and allowed the borrowing institution to return to normal operations. Additional policy responses for the benefit of the AHEL borrower followed for 21 of those firms. In the remaining case, Brazil’s privately owned deposit insurer’s AHEL support (and supplemental guarantees) of BTG Pactual in 2015—following its CEO’s legal imbroglio unrelated to the health of the bank—still funded a substantial balance sheet restructuring, with BTG halving its balance sheet size by the end of 2016.¹⁹ The most common additional intervention was a capital injection, which occurred in 13 of the cases—and typically alongside other interventions. Four of the AHEL programs facilitated mergers with healthier institutions. Two functioned as short-term bridges to more controlled failures of the borrower, and two to government takeovers. Several other parallel policy measures were common, such as various forms of asset or liability guarantees and government-led restructurings. Often, policymakers left AHEL facilities in place longer than initially intended so the liquidity could continue to buttress these other policy measures²⁰—particularly if

¹⁸ The Indonesian authorities’ use of an ostensible BBEL facility to provide AHEL assistance to Bank Century is demonstrative of financial authorities’ general preference for providing liquidity assistance through standing, or other BBEL, facilities. See Indonesia–Bank Century.

¹⁹ Moreover, the Brazilian central bank followed the deposit insurer’s AHEL intervention by exempting emergency liquidity assistance from reserve requirements.

²⁰ If the AHEL facility becomes a resolution financing facility, policymakers often supplement it with a fiscal guarantee; see, for example, Arnold (2025a).

initial attempts at the structural policy measures were inadequate, such as too small a capital injection.

AHEL assistance measures are often the first response of financial policymakers given the relative bureaucratic ease of making loans (typically through the central bank) versus deploying other policy measures (normally involving legislation or negotiations with private parties) and can thus be an effective bridge to other stabilization policies.

The consistency of policymakers' needing to implement additional measures across these cases should disabuse future crisis-fighters of the notion that an institution being run on ever has "just a liquidity problem"; the need for an AHEL program is itself indicative of the market's assessing the firm to have a viability problem.²¹

Greater up-front recognition that additional policy measures will be necessary should enable policymakers to speed up their disbursement of AHEL assistance even further; they need not worry about getting all the policy trade-offs right—for example, moral hazard, encouraging the return to private markets, etc.—in the terms of the AHEL assistance. Longer-term incentives for the borrower can be managed by the terms of the subsequent interventions. That ex ante recognition that more actions will be necessary should also encourage crisis-fighters to be mindful to immediately use the breathing room granted by an AHEL program to craft the more structural additional policy measures.

The long-term use of AHEL support as the primary or only intervention can result in increased risk to the lender, the creation of adverse incentives for bank creditors, and legal or political ramifications. In the case of the Central Bank of Cyprus's (CBC's) lending to Laiki Bank, the extended provision of ELA (in the absence of any bank holiday or withdrawal suspensions) allowed uninsured depositors to continue to run on the bank; by September 2012—roughly 11 months after the ELA began—the ELA had effectively replaced deposit funding. Given that uninsured deposits made up roughly half of Laiki's deposit base around the time it started receiving assistance, this runoff greatly reduced the amount of bail-in-able debt in Laiki's ultimate resolution. Further, since no other structural interventions had been implemented, the ultimate suspension of ELA (at the decision of the ECB Governing Council) would have—in the absence of an external aid package—resulted in the bankruptcy of Laiki. It was therefore the reliance on ELA and failure to implement other interventions that forced Cypriot authorities to obtain an external aid package. In Ireland, scholars have criticized the extended supply of ELA for four years as being used to delay liquidation of Anglo Irish Bank. The Central Bank of Ireland's (CBIR's) ELA provision ultimately ran up against European Union (EU)-level legal prohibitions, which forced the Irish authorities' hands in completing a delayed resolution. Aware of similar risks, and in contrast to the later cases of Ireland and Cyprus, United States authorities deliberately—and privately—shortened the maturity,

²¹ This conclusion is consistent with the large, recent study of 160 years of panel data on US banks from Correia, Luck, and Verner (2024). That study concludes that "bank runs can be rejected as a plausible cause of failure for most failures in the history of the U.S. and are most commonly a consequence of imminent failure." Using data for 46 countries dating back to 1870, Baron, Verner, and Xiong (2021) similarly finds that "panics tend to be preceded by large bank equity declines, suggesting that panics are the result, rather than the cause, of earlier bank losses."

relative to the duration authorized, of their AHEL loan to Bear Stearns to force Bear's management to accept a more permanent and structural solution to Bear's challenges (an acquisition).²²

3. Legal Authority: What statutory and regulatory provisions framed the AHEL lender's intervention?

In general, officials faced two primary legal restrictions on their ability to implement AHEL programs. First, policymakers generally had to invoke special, emergency authorities—which tend to carry higher invocation thresholds than standard lending authorities—whether it was to lend to nonbanks or to lend to banks on nonstandard terms, such as lending against collateral not accepted by standing liquidity facilities.

Second, central bank lenders were constrained in their lending by laws and regulations stipulating fully secured lending or, in some cases, solvency constraints. Even in cases where central banks have shifted their financial assessment of potential borrowers to focus on their viability, the relevant regulations and statutes tend to still focus on solvency (BIS CGFS 2017; Dobler et al. 2016).

In the event, many central banks interpreted that these requirements could be satisfied by protecting their balance sheets through broader means than collateral policies—for example, by obtaining loss-sharing arrangements, guarantees, or other recourse on their lending (see Key Design Decision No. 10, Balance Sheet Protection). Two exceptions to this rule were noteworthy. First, the governing statute of the Central Bank of the Russian Federation (CBR) permitted fully unsecured lending (though the CBR did not expressly invoke this authority in the Russia–Otkritie case). Second, a provision in the governing statute of the Central Bank of Iceland (CBI) allowed the CBI, in systemic scenarios, to deviate from standard collateral requirements. Strict solvency requirements on AHEL lenders can prove difficult in practice, as both the conditions and exigency of a crisis can make point-in-time solvency difficult to assess; see, for example, Dobler et al. (2016) and Goodhart (1999).

In the EU, Article 123 of the Treaty on the Functioning of the European Union (TFEU), prohibits monetary financing, which European regulators have interpreted to include central bank lending to insolvent institutions. In the Eurosystem, ELA is carried out by NCBs—the central banks of the member states of the EU that comprise the Eurosystem. As such, any ELA extended by an NCB (or by non-Eurosystem EU central banks) to an insolvent bank would be a violation of primary (foundational) EU treaty law and is thus legally prohibited. Notably, this proscription applies to EU members wholly, inclusive of those that have not adopted the euro. As a result, in the surveyed AHEL interventions, EU central banks

²² If authorities had shortened the AHEL program's maturity *publicly*, the revision could have backfired if Bear's acquisition had not consummated by the new deadline. As a Bank for International Settlements (BIS) working group on central bank liquidity assistance later writes, generally, "If the solvency of the recipient deteriorates after the receipt of [liquidity assistance], termination of assistance or measures to reduce counterparty risk (eg by shortening the duration of [liquidity assistance] or demanding additional collateral) can be problematic. They can precipitate the failure of the firm or hinder the process of dealing with its failure, unless other authorities have the ability to provide funding in a timely manner" (BIS CGFS 2017).

frequently obtained a guarantee from the fiscal authority when the solvency of their counterparties was in question. This legal prohibition under Article 123 then resulted in additional constraints on the central banks' AHEL program sizes, given that the binding constraint was often the fiscal guarantee they could obtain. (See Key Design Decision No. 7, Source and Size of Funding; Ireland–Anglo Irish, Spain–CCM, and UK–Northern Rock. For more on the application of TFEU Article 123 on EU central bank ELA, see Arnold [2025a].)

Likewise, outside the EU, central bank AHEL programs often ran up against the hard limits of the law with respect to lending “fully secured” or only to “solvent” institutions. Pushed far out on their risk curves but limited by those legal standards, central bank lenders found means aside from collateral stipulations to satisfy legal requirements, including, but not limited to, EU-style fiscal guarantees (see Key Design Decision No. 10, Balance Sheet Protection). In Switzerland, the Swiss National Bank's (SNB's) lending was not strictly solvency-restricted, but rather was restricted to lending against “sufficient collateral.” In the face of limited acceptable collateral, the SNB met this requirement in its first AHEL facility (what it called “ELA+”) by taking preferential bankruptcy rights. In its second AHEL facility, the public liquidity backstop (PLB), the SNB obtained preferential bankruptcy rights *and* a state guarantee.²³

In Indonesia, the central bank's solvency limitations were self-prescribed; facing a crisis that Bank Indonesia officials said was systemic, Bank Indonesia reduced its capital adequacy ratio requirements and collateral requirements when lending to Bank Century.²⁴

Non–central bank lenders (NCBLs) were considerably less legally constrained in their ability to provide AHEL assistance. However, some limitations were binding. In Brazil, the Credit Guarantee Fund (Fundo Garantidor de Créditos, FGC)—the private (but closely supervised) deposit insurer that provided the AHEL facility—was limited in the size of its AHEL lending by regulations passed by an interministerial council (which included the central bank and ministry of finance). When the FGC lent to BTG Pactual, the size of the lending relative to the FGC's balance sheet required that the FGC obtain a systemic risk determination from the Central Bank of Brazil.

4. Administration: What administrative process and infrastructure supported the AHEL intervention?

Most central banks administered AHEL programs through existing structures, in cases where they publicly specified how the program was administered. However, in some notable exceptions, they set up new lending infrastructures. Both methods came with benefits and challenges.

²³ In contrast to SNB lending writ large, the emergency ordinance that authorized the PLB facility and its state guarantee also came with the legal requirement that Credit Suisse's solvency be verified by the Swiss bank supervisor. This solvency requirement is standard in broad-based lending, though not legally required.

²⁴ This action, though, was highly controversial. Numerous lawsuits, government investigations, and prosecutions followed Bank Indonesia's handling of the Bank Century case.

Existing Administrative Infrastructure

Administration through existing facilities is most often accomplished through the standing discount window administrative infrastructure. Another option is to use the infrastructure of any previously erected BBEL facilities. For instance, in the US–Lehman LBI case, the Federal Reserve applied LBI-specific modifications to a BBEL facility that the Fed had stood up earlier in the crisis.

The use of preexisting administrative infrastructure often enabled quicker implementation, less administrative work, and lower operational risk. In one case, the central bank creatively used an existing facility for commercial banks to on-lend to an investment bank: The Federal Reserve used its existing discount window infrastructure to lend indirectly to Bear Stearns, an investment bank, via an on-lending arrangement with JPMorgan. While the Fed had the legal authority to lend to Bear Stearns directly, using the existing discount window administrative setup was operationally expedient. Thus, it created the back-to-back structure to lend through commercial bank JPMorgan. This allowed the Fed to make the decision and loan within “the space of a few hours” and take advantage of JPMorgan’s existing familiarity with Bear Stearns’s collateral; JPMorgan had been Bear Stearns’s clearing bank in the triparty repo market—which reduced some of the uncertainty around collateral valuation.

Intragovernmental Coordination

Being able to leverage existing structures may require intragovernmental coordination. In Latvia, the ministry of finance provided Parex with term deposits backed by government debt securities, allowing Parex Bank to then use those securities as collateral to borrow from the central bank. This arrangement meant that the central bank was not required to create new facilities or modify existing facilities’ collateral terms. This decision also shifted the counterparty risk of the borrowing from the central bank to the fiscal authority, a decision that was ex post validated in part by Parex’s insolvency and subsequent recapitalization and restructuring—much of the emergency liquidity was later converted into equity.

Conversely, a lack of intragovernmental coordination may hinder administrative ease. As former Federal Reserve Bank of New York (FRBNY) officials later noted, Bear Stearns’s primary regulator—the Securities and Exchange Commission (SEC)—was not particularly supportive or prepared in handling the near-collapse of Bear Stearns, for which the Fed provided an AHEL loan. This shortcoming added difficulty to the already challenging administration of an AHEL program, as the Fed was operating in an information deficit because it was not Bear’s primary regulator. Although the rescue operation was ultimately implemented in time, deeper information-sharing and collaboration between the Fed and SEC likely would have lessened the burden on an already stretched central bank.

New Administrative Infrastructure

In some cases, central banks stood up entirely new facilities to administer the AHEL facility. The administrative design of new facilities was often informed by existing BBEL or AHEL facilities, which could reduce the time and resources needed for administrative setup.

While most of the new facilities were administratively straightforward in mirroring discount window facilities, the Bank of England's (BoE's) creation of off-balance-sheet trusts—to provide AHEL assistance, in the form of collateral swaps, to HBOS and RBS—stood out from the others. This structure made use of newly established trusts to hold the collateral posted by HBOS and RBS. The trust structure enabled the Bank of England to avoid technically taking ownership of the collateral (despite having a beneficial interest in it)—which would have otherwise required the borrowers to publicly register the BoE's charge on that collateral (Plenderleith 2012). Administering the AHEL assistance through these trust structures thus contributed to the Bank of England's ability to keep this assistance confidential; see Key Design Decision No. 6, Communication and Disclosure.

Often, crisis-fighters need to implement a policy response after a similar private sector effort fails to come to fruition. Those failed efforts may still be useful for easing the administrative workload of the ultimate intervention. For instance, when a consortium of large financial institutions was forced to abandon a private rescue of AIG because of the fallout from the failure of Lehman Brothers, the Fed used the abandoned private sector term sheet as the skeleton of its Revolving Credit Facility (RCF) for the insurance giant AIG. However, the original terms of the AIG RCF facility proved too onerous, forcing policymakers to revise them later; see Key Design Decisions No. 8, Rates and Fees, and No. 9, Loan Duration. As the US-AIG RCF case shows, policymakers leveraging preexisting private sector administration efforts should take care to modify commercial terms to serve the appropriate public policy goals.

Although this survey does not examine stand-alone financing-in-resolution frameworks, the central banks in our surveyed case studies commonly novated the AHEL facility to an authority tasked with resolution financing, consistent with practices in their jurisdictions; see Key Design Decision No. 10, Balance Sheet Protection.

Outside Firms

In some cases, central banks made extensive use of outside experts for valuation, accounting, and advisory services. At times, these services were integral to the disbursement of liquidity. The use of credible external consultants can provide greater legitimacy and expedience to an AHEL facility. For example, for the AIG RCF, the FRBNY contracted outside accountants to simulate stress scenarios on AIG daily and report those outcomes to FRBNY leadership—after which it would decide whether or not it was still comfortable lending (Dahlgren 2018). Moreover, the FRBNY hired outside consultants to provide collateral valuation services in part based on those consultants' insurance expertise, since the Fed was not accustomed to lending to insurance companies. Likewise, in Denmark, the central bank engaged PricewaterhouseCoopers to verify the cash flow statements from Roskilde Bank—particularly related to the behavioral conditions of the AHEL facility—that were required for Roskilde to draw on the AHEL facility. However, authorities should also be wary of potential conflicts of interest in their choice of outsiders, such as by avoiding, where possible, external firms that themselves have a stake in the rescue.

5. Governance: Which bodies governed or otherwise provided oversight to the AHEL intervention? What forms did the oversight take?

Nearly all the cases examined featured standard disclosure processes and forms of accountability characteristic of crisis-time interventions. For example, all recipient banks filed annual or quarterly reports disclosing receipt of AHEL assistance. Following an intervention, investigative commissions including parliamentary bodies, investigative committees, government auditors general (and, in some controversial cases, courts and prosecutors) frequently audited the government's actions in providing AHEL assistance. After several of the interventions studied here, legislative bodies stood up special committees charged with investigating and reporting on the AHEL programs.

Legal requirements commonly mandated central bank lenders to obtain approval for their actions from other bodies of government. For instance, in the UK, the BoE Tripartite Agreement with the Financial Services Authority and Her Majesty's Treasury mandated that the governor of the BoE obtain authorization from the chancellor of the Exchequer (Treasury) to provide any AHEL support.²⁵

Fiscal guarantees, common in AHEL cases, generally increased the governance responsibilities of other government bodies with respect to the AHEL program. In Denmark, the treasury guaranteed the Danmarks Nationalbank's (DNB's) AHEL lending to Roskilde; the treasury, in turn, required legislation from parliament, and the process took nearly two months. The Spanish treasury's guarantee of the Bank of Spain's lending—legally required to execute the AHEL support of CCM—required an emergency executive ordinance with ex post parliamentary approval. Similarly, both the SNB's AHEL facilities for Credit Suisse required new legislation, in whose place an emergency executive ordinance stood. While that emergency ordinance required and later obtained approval in the upper house of parliament, the lower house voted against authorizing the ad hoc PLB facility for Credit Suisse. Since the SNB had already committed the amounts authorized by the ordinance, the vote had no impact, and it was debated under the understanding that it was nonbinding.

AHEL programs were sometimes revised by other government bodies, even if already executed. In Sweden, the central bank extended AHEL assistance to Carnegie Investment Bank AB; when parliament passed new legislation, the new laws mandated that AHEL programs be provided by the National Debt Office (NDO), so the central bank was required to novate the Carnegie AHEL line to the NDO.

In Iceland, decisions surrounding the extension of emergency liquidity to Kaupthing were made informally between the CBI's board members and the prime minister; years later, parliament passed legislation requiring that decisions on emergency loans be made during

²⁵ While not in force for the Global Financial Crisis (GFC)–era US case studies in this series, the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, in part in response to the surveyed US AHEL programs, mandated the Fed's Section 13(3) emergency lending authority obtain prior authorization from the Treasury secretary—which had been standard practice previously anyway. (The legislation also prohibited lending to individual firms.)

meetings that included at least the CBI’s governor and three deputy governors—and that the decision must be documented in meeting minutes.

Governance and oversight of AHEL programs were also at times a binding constraint in cases of non-central bank lenders. In Brazil, for example, the private deposit insurer required a systemic risk determination from the central bank to provide an AHEL program for BTG Pactual when its lending reached a certain size threshold. In Latvia, the fiscal authority required authorization from the Latvian executive cabinet to provide its AHEL program for Parex.

ECB-NCB Governance in the Eurosystem: A Special Case

In the eurozone specifically, central bank AHEL programs are implemented under the region’s separate ELA protocol. ELA is administered by respective national central banks—typically when the relevant institution or its collateral is not eligible for further ECB financing. However, under Article 14.4 of the Statute of the European System of Central Banks and the European Central Bank, the ECB Governing Council retains the authority to prohibit or cap the amount of ELA if it “considers that these [ELA] operations interfere with the objectives and tasks of the Eurosystem”—which can hinder a national central bank’s desired provision of AHEL assistance (ECB 2013).

For example, when Anglo Irish Bank ran out of ECB-eligible collateral, the finance ministry recapitalized it with government debt securities—which were then used as collateral at the CBIR. To members of the ECB Governing Council, this intervention could have been interpreted as monetary financing of the traditional variety, in which the monetary authority finances the government through reserve creation. Similarly, if the government bonds were low quality and thus could not be considered to adequately plug a capital hole, the CBIR lending could have been construed to be to an insolvent firm—that is, the EU-specific additional variety of “monetary financing.” As such, then-ECB President Jean-Claude Trichet explicitly conditioned the CBIR’s AHEL assistance to Anglo on structural reforms, fiscal consolidation, and external aid programs, due to the ECB’s concerns about Anglo’s solvency and about Irish authorities’ attempts to resolve those solvency concerns with government securities. Trichet noted that, generally, it would be imperative for the ECB to intervene if there were concerns about “monetary financing” or impairment of monetary policy transmission (emphasis added; italics in original):²⁶

The provision of *Emergency Liquidity Assistance (ELA)* by the Central Bank of Ireland, as by any other national central bank of the Eurosystem, is closely monitored by the Governing Council of the [ECB] as it may interfere with the objectives and tasks of the Eurosystem and **may contravene the prohibition on monetary financing**. Therefore, whenever ELA is provided in significant amounts, the Governing Council needs to assess whether it is appropriate to impose specific conditions in order to protect the integrity of our monetary policy. In addition, **in order to ensure compliance with the**

²⁶ In the EU, central bank lending to insolvent institutions is considered subject to Article 123 of the TFEU, which prohibits monetary financing. See Key Design Decision No. 3, Legal Authority; see also Arnold (2025a).

prohibition on monetary financing, it is essential to ensure that ELA recipient institutions continue to be solvent. (Trichet 2010)

Likewise, in the case of Laiki Bank, the Cypriot finance ministry recapitalized the bank with an unfunded sovereign bond. Eventually, in 2013, after rating agencies had downgraded Cyprus's credit rating, the ECB refused to approve further ELA from the Central Bank of Cyprus to Laiki. The ECB explicitly conditioned any further ELA approvals on the successful implementation of a support package from the EU and International Monetary Fund (IMF) that adequately secured the solvency of the potential recipient banks.

Monetary financing and monetary policy considerations aside, all EU AHEL programs—regardless of the country's membership in the Eurosystem, and regardless of the funding's being provided by a central bank (so long as provided by an organ of government)—were subject to European Commission (EC) State Aid investigations, which could enforce binding constraints. In Latvia, the EC very nearly limited continuing AHEL assistance to a bank in resolution on the basis of incompatibility with State Aid regulations. It ultimately allowed the program, but the EC's ongoing permitting of the AHEL assistance was subject to behavioral restrictions (see Key Design Decision No. 12, Other Conditions).

6. Communication and Disclosure: To what degree did crisis-fighters communicate and disclose details about the AHEL intervention? What was the tenor of the communications?

It is widely accepted in practice and in academic literature that emergency lending can benefit from being kept confidential, particularly the specific borrower details.²⁷ Such confidentiality—which should not be conflated with a lack of governance process over the confidentiality and publishing of such information—can prevent a market panic and allow the borrowing institution to continue operating under orderly conditions. As a BIS working group report on central bank liquidity assistance says, “By adopting measures to ensure ex ante transparency, external oversight during [a liquidity assistance] operation and ex post transparency, central banks ought to be able to ensure sufficient accountability to leave scope for a temporary reduction in real-time transparency where financial stability would otherwise be threatened” (BIS CGFS 2017).

However, the option for confidentiality typically doesn't exist in the case of an AHEL intervention; given the need for an ad hoc solution, the market is usually already focused on the institution's troubles. Market participants could suspect central bank liquidity assistance from suddenly altered market trading patterns, such as the suspected borrower's no longer bidding for funding in private funding markets or borrowing with upgraded collateral suggestive of its receiving a central bank collateral swap (BIS CGFS 2017; Wessel 2010, 120). Central bank activity visible to market participants may also suggest an active emergency

²⁷ See, for example, Alvarez and Baxter (2011), Anbil and Vossmeier (2021), Armantier et al. (2015) Bernanke (2015b), Gorton and Ordoñez (2020), Gorton and Tallman (2018), Kelly (2024c), Wiggins, Fulmer, et al. (2023), and the references therein.

liquidity program, such as if the central bank sourced foreign exchange or other collateral in private markets to fund the assistance (BIS CGFS 2017).

Thus, in a typical AHEL provision, the communication incentives are flipped: Policymakers should endeavor to quickly communicate as large a response as is both possible and credible. Failing to do so leaves policy benefits on the table, as markets are aware of the institution's problems but unaware of the AHEL response or all its details.

In the Case of Transparency

As noted, AHEL interventions are typically needed in cases where the institution's troubles are already widely understood (or perhaps inflated) by the market. As such, crisis-fighters should leverage communicating with the market to project as much calm as possible. Particularly, when true and credible, officials should endeavor to communicate that it is safe to continue to do business with the affected firm; officials should also encourage public proclamations of business-as-usual from the borrower's largest counterparties, as that may support market confidence (ECDC 2023); see, for instance, US–Bear Stearns, US–Lehman LBI, and US–First Republic.

Importantly, an AHEL program's announcement should be timed after the agreement is already in place; implying the need for an AHEL intervention before it is in place may exacerbate any panic. To the extent possible, the communication should also note the AHEL intervention's adequacy or ability to meet the bank's potential near-term obligations; if it falls short of that size, it may make sense to not mention any specific cap—while still noting the AHEL program's sufficiency. For instance, the Fed's limited communications and undue vagueness about its level of support to Bear Stearns—which, in the event, would only have been limited by available collateral—led to confusion in the market about the Fed's level of liquidity support, and even whether or not it was resolution financing (Cohan 2009, 70–71). See Key Design Decision No. 7, Source and Size of Funding, for a further discussion of sizing AHELs.

While, as discussed in the Introduction, an AHEL program itself is unlikely to contain a run, these communication measures may be at least directionally helpful, particularly alongside any communication that more policy measures are coming. Any impending responses should also be communicated alongside the AHEL intervention to the extent possible, to provide greater market confidence in the authorities' commitment to restore the viability of the borrowing institution. Moreover, AHEL communications can be helpful for limiting contagion if they relay a willingness to repeat the policy intervention as necessary—as in the cases of First Republic and Citigroup (Citi).

Before the Global Financial Crisis (GFC), to avoid moral hazard, central bankers often advocated “constructive ambiguity” in their communications about their willingness to support troubled financial institutions. Since the GFC, central bankers have increasingly distanced themselves from that strategy. To prevent contagion with an AHEL intervention, central bankers should indeed replace constructive ambiguity with constructive clarity about their readiness to repeat such support in the future if warranted and the legal

conditions are met. As Reserve Bank of Australia Assistant Governor Brad Jones said regarding the superiority of providing greater clarity on the availability and terms of emergency liquidity, the “ad hoc approach of yesteryear was unhelpful for decision makers and could exacerbate market uncertainty” (Jones 2023).²⁸

During the early days of the Banking Crisis of 2023, there was some renewed market volatility—particularly for First Republic, which had received AHEL assistance—following remarks from US Treasury Secretary Janet Yellen before one chamber of the legislature (Bautzer and Prentice 2023; Wilkie 2023a; Yellen 2023a). However, she revised her testimony for the other chamber the next day—to market-calming effect—to include that the crisis-fighting tools already implemented were “tools we could use again,” adding, “certainly, we would be prepared to take additional actions if warranted” (Wilkie 2023b; Yellen 2023b).

Similarly, in announcing its assistance to Northern Rock, the Bank of England said it stood “ready to make available facilities in comparable circumstances, where institutions face short-term liquidity difficulties” (BoE 2007). When providing a relief package inclusive of a contingent loan to Citi in 2008, US policymakers signaled repeatability in the press release, saying that the government would “continue to use all of our resources to preserve the strength of our banking institutions and promote the process of repair and recovery and to manage risks” (Fed, FDIC, and Treasury 2008). Just a few months later, policymakers announced a nearly identical ad hoc contingent liquidity facility for Bank of America (BofA), drawing on their experience in creating such a facility for Citi.

However, there are risks to signaling future support. Such signals (a) may not be credible or (b) may implicate other specific institutions. Both issues were triggered in the case of the Fed’s AHEL assistance to Bear Stearns—the first indirectly and the second directly:

- *On credibility.* The Fed Board’s confidential approval for the AHEL assistance to Bear technically included authorization for the Fed to provide such assistance to other broker-dealers as well (Fed 2008a). However, the Fed did not publicly communicate this fact. If it had, such communication may have backfired, as the Fed ultimately thought itself unable to lend to Lehman Brothers six months later. At the time of the Bear rescue, the Fed’s public communications did not provide any clarity about the potential terms of future rescues. Still, over the ensuing months, the Fed contended with overconfidence in the market that similar assistance would emerge for Lehman (Bernanke 2015a, 289–90).
- *On avoiding implicating specific institutions.* At least part of the reason the Fed’s AHEL assistance to Bear was needed was that the Fed had recently announced a broad-based program that would be available to large dealers such as Bear. Though

²⁸ He added that constructive clarity can support resolution and recovery planning (see also Arda and Nolte [2025]) and that constructive ambiguity is less necessary as a moral hazard cure given post-GFC “tighter prudential liquidity requirements and severe, well-telegraphed consequences for the leadership of (and equity holders in) mismanaged institutions that require public support” (Jones 2023).

this program was broadly based, some in the market took its introduction as a statement about the health of Bear, and such concerns grew (Baxter 2018; Cohan 2009, 25; GAO 2011a).

In the Case of Confidentiality

If it is possible to keep the borrower's need for AHEL assistance confidential, doing so may be desirable for the aforementioned benefits of avoiding any additional instability—typically only in the case where the market does not know the extent of the borrower's financial distress. However, such confidentiality comes with the risk of being discovered by the market, particularly if such lending can be deduced either in the market or from regular central bank publications of its balance sheet data. For instance, the Basel Committee on Banking Supervision reported that, even prior to receiving ad hoc assistance, Credit Suisse feared tapping standing facilities out of concern that market participants would have been able to infer the firm's borrowing “from aggregate balance sheet positions in the Swiss National Bank's regular reporting with some time delay, which might have further deteriorated their confidence in the bank” (BCBS 2024).

An AHEL lender taking the risk of attempting to keep the lending confidential should be prepared for such a discovery with appropriate communication measures and additional policy measures if necessary (Plenderleith 2012). **Moreover, central banks should have confidentiality governance measures—a clear set of rules about its confidentiality procedures—in place ex ante, or leaks may also damage the government's credibility in crisis, as in the case of Northern Rock.**

The run on Northern Rock accelerated when news of its request for AHEL assistance from the Bank of England leaked before any press release—and before the assistance was in place. By contrast, the Swiss National Bank announced the AHEL assistance available to Credit Suisse two days after the bank had begun borrowing. The two-day wait may have risked leaks or unnecessarily added to uncertainty; however, the SNB was also able to then announce the AHEL assistance alongside other measures, inclusive of CS's merger with UBS.

In a postmortem on Northern Rock, the UK House of Commons's Treasury Committee called the BoE's failure to quickly finalize and announce the AHEL intervention before the leak combined with its failure to swiftly and adequately respond to the press interest, “the worst of both worlds.” The committee added that, while markets may have responded negatively to an AHEL intervention regardless, an official communication before the leak could have provided “a reasonable prospect that the announcement would have reassured depositors rather than having the opposite effect” (House of Commons 2008).

Similarly, for both Bank Indonesia's liquidity assistance to Bank Century and the National Bank of Moldova's (NBM's) assistance to a consortium of banks, markets had determined that the institutions were facing runs before the preexisting AHEL assistance was made

public.²⁹ The liquidity assistance could not calm markets, as the markets were unaware of its existence.

By contrast, the BoE was able to keep its AHEL assistance to RBS and HBOS largely confidential until the acute phase of the crisis had passed.³⁰ However, this was accomplished through two design features and one circumstance of the crisis (Plenderleith 2012):

- Design feature: The programs were structured as collateral swaps, so they did not increase the size of the BoE's balance sheet, which was regularly reported publicly.
- Design feature: The BoE implemented these programs through off-balance-sheet trust structures, which allowed the borrowers to avoid publicly registering the BoE's charge on those assets.
- Circumstance: Because these AHEL interventions occurred in the autumn of 2008 amid several similar broad-based programs, their presence was harder for market participants to discern.

Broadly, other lending programs or asset purchases may help mask AHEL activity depending on how the relevant central bank publishes its balance sheet data. Special purpose vehicles or other off-balance-sheet structures may not provide confidentiality if markets can parse the data on central bank balance sheet size (the central bank typically still has to fund the separate vehicle). If AHEL data is masked by being combined only with other broad-based lending, the effectiveness of that confidentiality might depend on how much those facilities lack stigma.

Moreover, officials should not rely on other lending or asset purchases to mask an AHEL facility, as AHEL assistance may be implemented as the first intervention of a crisis—that is, before other policy measures—or amid a monetary policy environment of a stable or shrinking balance sheet, as in the US Banking Crisis of 2023 (see, for example, Kelly [2024c]).

Thus, authorities should consider whether categorizations in any regular balance sheet publications—or even the publication thereof—provide the right public interest balance

²⁹ In both cases, the central banks eventually publicized their support after it became untenable to maintain secrecy (in the Indonesian case because Bank Indonesia recognized that public support for Bank Century would help the institution, and in the Moldovan case because of a leak).

³⁰ In addition to the listed reasons that confidentiality was accomplished, the BoE was also able to avoid the news's leaking, which may have been assisted by the limited number of BoE officials that were aware of the AHEL support (see UK-HBOS and RBS). A BIS working group later commented on the general practice of keeping awareness of liquidity support operations to a limited number of officials, noting that while it may help limit inadvertent and deleterious disclosure of such operations, it also makes "more challenging the external oversight of the central bank during this time. A number of countries responding to the challenges have put alternative arrangements in place to enable effective external oversight of [liquidity assistance] operations still to take place. This response is particularly important if a central bank is lending outside its published framework" (BIS CGFS 2017).

between disclosure and opacity. For an analysis of this issue in the US, see Kelly (2024c), and in the UK, see Plenderleith (2012). As the BIS working group noted:

In recent years, there have been more channels through which information about [a liquidity assistance] operation may find its way into the public domain (intentionally or not) even in situations where the central bank may judge this to be harmful to financial stability. This is in part due to an ongoing trend towards enhanced transparency of central bank operations themselves. (BIS CGFS 2017)³¹

7. Source and Size of Funding: How did the AHEL lender decide on the size of the loan? Where did it source the funds?

Sizing an AHEL Program

When possible, authorities should seek to deliberately size an AHEL program to be at least large enough to cover all potential outflows over the intended timeframe for the assistance. Announcing such sizing can at least eliminate the “first-mover advantage” for short-term counterparties of the institution that could run. For instance, during the Banking Crisis of 2023, several banks sought to preempt any potential run contagion by putting out statements to the market to demonstrate they had sufficient access to liquidity—inclusive of available Fed and Federal Home Loan Bank (FHLB) liquidity—to cover the potential outflow of 100% of uninsured deposits—to varying degrees of success.³² As mentioned in the Introduction, this approach should not be relied upon as a sufficient condition to wholly stop or reverse a run.

Denmark’s central bank explicitly said it would “provide the necessary liquidity” to Roskilde in 2008 when it established an unlimited and unsecured liquidity facility; yet, Roskilde would still need to tap the facility for years, until its final restructuring. In the Bear Stearns case, the Fed did not announce any specific cap on its bridge loan to Bear, and Bear’s press release announced the bridge loan would allow it “to access liquidity as needed” (Bear 2008; Fed 2008b). Yet, Bear’s financial deterioration accelerated the one day the bridge loan was

³¹ The report went on:

Another reason relates to the practical implications of domestic and international initiatives aiming to increase market transparency, which oblige a firm to disclose more detailed and more frequent balance sheet information or the extent of its asset encumbrance that may reveal the existence of [tapping a liquidity assistance program]. . . .

Firms are subject to regular financial reporting requirements. A firm receiving [liquidity assistance] is not likely to be able simply to stop making such disclosures that can potentially reveal [a liquidity assistance] operation. It would be preferable instead to ensure ex ante that these regular disclosures are made in a format which does not put financial stability at risk through premature revelation [of the liquidity support]. (BIS CGFS 2017)

³² See, for example, Bank of Hawaii Corporation (2023), East West Bancorp (2023), First Citizens BancShares (2023), Metropolitan Commercial Bank (2023), PacWest Bancorp (2023a), PacWest Bancorp (2023b), and Presence Bank (2023). Notably, UMB Financial Corporation explicitly reported having just 52% coverage, inclusive of other collateralized lending sources (UMB Financial Corporation 2023). On the several occurrences of 100% liquidity coverage of uninsured deposits in the Banking Crisis of 2023 failing to prevent additional outflows, see Kelly and Rose (2025).

in effect. (The continued run in this case may have been at least partly due to concerns over adequate collateral, since the bridge loan was a secured lending facility; MarketWatch cited a Fed staffer saying that the size of the loan wasn't certain and would depend on Bear's credit needs and available collateral [Robb 2008].)

In the case of Credit Suisse, officials sized the AHEL assistance, at 200 billion Swiss francs (CHF; USD 217 billion)³³ to cover “virtually all short-term liabilities of the bank” (SNB 2023). Yet, outflows of deposits and client assets continued in the weeks that followed; after more than a month, outflows had moderated but not yet ceased (CS 2023). This further underscores that crisis-fighters should not expect an AHEL program to “work” in isolation just because market participants can see that it is large enough for the borrower to meet all possible outflows and avoid a technical default; additional policy responses to restore the firm's viability will almost certainly be necessary.

In practice, the sizing of the AHEL program may be limited by the source of funds, which we discuss next.

Source of Funds: Central Banks

In 18 of the 22 cases surveyed, the central bank was the AHEL lender—at least initially. Central bank reserve creation is the most unconstrained form of funding, though it may in practice be limited by the central bank's ability to get adequately secured, the lending's impact on monetary policy, or, in the case of being secured by a fiscal guarantee, by the finance ministry's fiscal space or fiscal authorization (see Key Design Decision No. 10, Balance Sheet Protection).

The central bank may also fund its AHEL interventions out of existing balance sheet assets, which would imply that the preexisting size of the balance sheet may function as a constraint. In the Kaupthing case, the CBI tapped its on-balance-sheet foreign exchange reserves, which ultimately proved a constraint on the size of the assistance. The CBI had insufficient foreign reserves despite its access to euro swap lines with Norway, Sweden, and Denmark (Hoffner 2023). Broadly, having more extensive access to foreign exchange ex ante is likely helpful, particularly for banks with cross-border operations. The SNB provided more than half of its AHEL assistance to Credit Suisse in foreign exchange, which it could buttress with its standing access to central bank swap lines and the Fed's foreign repurchase agreement (repo) facility (the latter of which it drew on materially) (Swiss Parliament 2024). In the UK, the BoE used its swap line with the Fed to provide AHEL dollar facilities to HBOS and RBS. However, some standing central bank swap arrangements come with restrictions on the downstream use of borrowed funds, which may limit a central bank's ability to use the foreign exchange for an AHEL facility (Wiggins, Hoffner, et al. 2023).

³³ Per the Bank for International Settlements, USD 1 = CHF 0.92 on December 30, 2022.

In case of Latvia's Parex, the fiscal authority first provided a de facto collateral swap, to allow Parex to obtain liquidity from the central bank's standing facilities.³⁴ The finance ministry provided government debt securities against collateral that the central bank could not accept (mostly loan portfolios), allowing Parex Bank to use those securities as collateral to borrow from the central bank. This arrangement meant that the central bank was not required to create new facilities or modify existing facilities' collateral terms, could avoid complex loan portfolio valuations, and effectively shifted the counterparty risk of the borrowing from the central bank to the fiscal authority (a decision that was validated in part by Parex's insolvency and subsequent recapitalization and restructuring; much of the emergency liquidity was later converted into equity). In the UK, the BoE borrowed short-term government debt securities from the Debt Management Office, which it then on-lent to HBOS and RBS in exchange for unsecuritized mortgages and loans.

Source of Funds: Non-Central Bank Lenders

In four cases surveyed, non-central bank lenders, public and private, were the initial source of AHEL funding, while central-bank-provided AHEL programs would also sometimes later novate to an NCBL. These NCBLs included fiscal authorities, private banks, deposit insurers, state-owned enterprises, debt management offices, and resolution authorities. The reasons for an NCBL to provide emergency liquidity varied. NCBL lending could serve to reassure markets or to reduce the central bank's counterparty risk. NCBL lending could also be an effective method to avoid legal or procedural limitations on central bank lending. In other cases, AHEL interventions were legally (or, as a matter of government practice) the responsibility of non-central bank government bodies.

In some cases, central banks intervened only after failed attempts by an NCBL to provide liquidity; this was particularly the case when there was the possibility of a fully private sector alternative. In Germany, two sets of liquidity facilities for IKB were provided: one from state-owned development bank KfW and other smaller facilities from private banks. In the US-AIG RCF case, the facility was modeled after a drafted private sector facility that the lending consortium abandoned when it could no longer provide the requisite financing in the aftermath of the Lehman failure. In Denmark, the central bank provided its liquidity facility, which was in turn guaranteed by a private deposit insurer, after the Danish Bankers Association confirmed that the banking sector "was unable to provide the necessary funds to Roskilde" (NAO 2009). In the UK, the Bank of England ultimately created AHEL facilities but had first planned to provide loss-sharing support to a private sector AHEL program for HBOS before it became apparent that the arrangement was unworkable. In Brazil since 2013, the financial industry's private deposit insurance organization provided emergency liquidity prior to the central bank, and the recipient institution would appeal to the central bank only when those liquidity measures were insufficient.

³⁴ In the Ireland-Anglo Irish and Cyprus-Laiki cases, the fiscal authority also issued government debt that could be used as collateral for central bank borrowing, but those issuances were ad hoc capital injections rather than exclusively about obtaining liquidity (see Ireland-Anglo Irish; Schaefer-Brown [2024]).

In the US in 2023, a consortium of 11 large private banks placed a USD 30 billion uninsured and unsecured deposit with First Republic. The public sector also coordinated this intervention, as the Treasury secretary first suggested the intervention to JPMorgan CEO Jamie Dimon. The consortium's action was motivated by the desire to signal confidence in the broader banking system, to help limit contagion. Notably, in this case, the typical order was reversed, with central bank lending coming first, followed by NCBL lending: First Republic had obtained liquidity from the Fed before it received the private sector AHEL support. The private sector's AHEL offered more generous terms than the Fed's and at first replaced Fed lending. However, First Republic continued to lose deposits in the ensuing weeks and gradually escalated its reliance on Fed lending despite the consortium deposit.

In some cases, the transfer of the administration of an AHEL facility to another body of government was part of the resolution or restructuring process. For instance, Denmark's resolution authority took over the AHEL facility from the central bank when Roskilde was restructured in a "good bank-bad bank" split. Similarly, UK's Treasury assumed the BoE AHEL facilities for Northern Rock roughly six months after it nationalized Northern Rock (the Treasury had indemnified the BoE on its lending on these facilities months before the nationalization). In doing so, HM Treasury refinanced the AHEL facilities, thereby assuming the counterparty risk while restructuring the bank. As then-BoE Governor Mervyn King later writes, "When the Bank of England lent to Northern Rock in 2007, it was possible to predict when the [lending] assistance would reach its maximum [as prescribed by the haircut-adjusted amount of Northern Rock's collateral]. The limit was duly reached on the date predicted and the government had to take over the financing of the bank and the associated credit risk" (King 2016, 205).

Relatedly, the central bank might hand the responsibility of providing an AHEL facility to an acquirer of the troubled institution. In the US, the Fed provided liquidity to Lehman Brothers's broker-dealer subsidiary (LBI), but once Barclays agreed to acquire the institution (though before it had done so), it took over the financing of LBI, thereby removing the Fed's counterparty risk (and marginally shrinking the Fed's balance sheet).

NCBL lending also faces criticism, particularly over its inherent constraints relative to central banks. For example, the IMF said that the private deposit insurer's provision of emergency liquidity in Brazil raised concerns, since the private insurer had no supervisory information about the institutions to which it was lending—and therefore lent with nonexistent or insufficient viability assessments; risked confidential data being leaked to the market; and would likely be unable to provide sufficient liquidity for large, systemic institutions (IMF 2018).

On the other hand, private sector NCBLs may be more agile lenders by being free of public sector restrictions on emergency lending, such as maximum durations or a minimum punitiveness mandated by statute or regulation. For instance, the bank consortium that assisted First Republic did so via an unsecured deposit; that is, it was free of collateral constraints that would have bound the Fed or the FHLBs. Yet, as the AHEL program size increases and the number of NCBLs involved grows (for example, the 11 banks involved in the First Republic intervention), the complexity of coordination increases, impairing the

efficiency often required in crisis contexts. Even if such coordination is successful, it may increase the risk of contagion if losses are realized or perceived on an AHEL loan. Sticking with the First Republic example, while all its deposits were ultimately protected by JPMorgan's bid for First Republic when it failed, other competing bids would have left the consortium deposit behind in resolution—likely subjecting the participating banks to losses.

At a certain size and complexity, such coordination would become impractical. As a result, NCBL AHEL facilities are likely practical for only relatively small interventions. NCBLs, whether private banks or quasi-public entities such as the FHLBs, are necessarily limited in the amount of liquidity they can provide by nature of lacking reserve-creation capacity or having capital constraints.³⁵ When fiscal guarantees from NCBLs are in place—usually as a result of large lending size, concern over solvency or viability, or legal limitations—the size of that NCBL guarantee necessarily limits the overall size of the AHEL assistance (see Key Design Decision No. 10, Balance Sheet Protection; Arnold (2025a); Ireland–Anglo Irish, Moldova–Consortium, Spain–CCM, and Switzerland–Credit Suisse).

8. Rates and Fees: How did AHEL lenders set the interest rate and fees on the AHEL assistance?

To this day, central bankers consistently use Walter Bagehot as a starting place for designing emergency lending operations; his 1873 dictum advises that, in a panic, central bankers should lend freely, against good collateral, and at a penalty rate (“a very high rate of interest”) (Bagehot 1873). While the penalty rate is pointed to as a means to discourage borrowing from the central bank and limit moral hazard (see, for example, [BoE 2007]), Bagehot's context was also that of a quasi-central bank (the private Bank of England of the time) whose ability to supply funds was limited by its gold reserves (Humphrey 1975).

A penalty rate may be appropriate for broad-based market facilities.³⁶ A Bagehot-esque, relatively high rate of interest can ensure that a broad-based lending facility functions as a “backstop”: naturally falling into disuse as market rates normalize from their period of stress. However, given that an AHEL facility is designed for the situation of a particular institution that is already the intense focus of the central bank and is receiving negative attention from the market, assessing relatively costly pricing to disincentivize use would be counterproductive (Brooks 2024). In such a case, usage will already be disincentivized by the market stigma of such borrowing and the heightened supervisory scrutiny of the draws on the AHEL line.

To be sure, Bagehot was writing about a systemwide spike in liquidity demand, not a question of viability at a particular institution. Where broad-based liquidity crises might be resolved by sufficient disbursement of emergency liquidity, we have discussed throughout this survey that AHEL programs are almost always only bridge solutions to more structural policy responses for individual institutions. As such, policymakers need not use the terms of

³⁵ On the FHLBs' limited liquidity access relative to the scope of the Banking Crisis of 2023 in the US, see, for example, Miao, Zuckerman, and Eisen (2023).

³⁶ For the sister YPFS survey of broad-based emergency liquidity programs, see Wiggins, Fulmer, et al. (2023).

the AHEL assistance to manage all of the potential moral hazard; they can manage moral hazard subsequently, subject to their level of confidence that other policy measures are forthcoming. Charging an interest rate on the AHEL assistance—inclusive of fees on undrawn amounts—that focuses on addressing all the moral hazard present in the rescue will likely lead to further deterioration of the borrower’s financial position—and market perceptions thereof. As IMF authors note, a too-high interest rate can “make survival of the institution (and repayment of [the lender]) less likely” and may be counterproductive to the AHEL lender (Dobler et al. 2016).³⁷

In several cases among those studied here, policymakers seemed to navigate these considerations well. For instance, in the Bear and Lehman interventions, both of which were intended from the outset to be bridge loans to ensure that those institutions could survive the acute phases of crisis, the Fed elected to lend at rates slightly lower than its own guidance for such loans—instead lending at its standing discount window rate (which was only a slight premium to the policy rate). However, for its conditional AHEL facilities for Citi and Bank of America, which came later in the crisis and were accompanied by other policy measures, the Fed applied a more traditional penalty rate—a benchmark funding rate plus 300 basis points (bps).

In 2023, then–Swiss National Bank Vice Chairman Martin Schlegel noted that pricing the SNB’s AHEL assistance to Credit Suisse required a “balance.” He said that while the SNB targeted a rate that was above the usual market rate, “it must not be so expensive that it becomes impossible for a bank to get out of its problems” (Fuster and Müller 2023). Moreover, a higher interest rate may have discouraged UBS from acquiring the foundering Credit Suisse. As such, AHEL lenders should make sure that any rate applied is sufficiently accommodative, flexible, or otherwise not a disincentive to other policy options that policymakers may be pursuing, such as a merger or a government takeover. Not only will AHEL lenders not want to preemptively foreclose other policy options, but it is also likely that the risk to the AHEL lender will fall following further policy measures, whether public or private solutions.

In an extreme case, the National Bank of Moldova deliberately lent at just 10 bps to three insolvent banks to avoid the systemic fallout from their uncontrolled failure. At the time, market rates were several hundred basis points. On its website, the NBM explicitly stated that it did so to avoid diminishing the banks’ resources available to pay out depositors. (The NBM’s loans were guaranteed by the state.)

³⁷ As a separate IMF report later puts it:

It is sometimes suggested that [emergency liquidity assistance] should be provided at ‘above market rates.’ However, there is unlikely to be a genuine market rate for an illiquid bank, or if liquidity stress is systemwide. Central banks need to strike a balance between providing incentives for a distressed institution to seek alternative funding, and moral hazard if [emergency liquidity assistance] were available too cheaply. The rate at which [emergency liquidity assistance] is provided (typically expressed as spread over a policy rate) should be sufficient to discourage use (for example, when there are genuine market alternatives), but not so high as to accentuate the strains the [emergency liquidity assistance] is seeking to alleviate. (Moretti, Chavarri, and Dobler 2020)

By contrast, in both cases surveyed that attempted to impose punitive rates as part of an AHEL program responding to an acute crisis—Latvia–Parex and US–AIG RCF—policymakers were forced later to amend these rates to arrest further financial deterioration at the borrowing institutions.

When lending to Parex in November 2008, the Latvian treasury charged an interest rate premium based on prevailing credit default swap spreads, leading to an extremely punitive interest rate. By December, officials lowered this rate to prevent an erosion of Parex’s capital ahead of an intended restructuring or liquidation—a justification with which the European Commission concurred. Later, when Parex had been restructured into a good bank–bad bank, officials charged a rate to the good bank that stepped up monthly, to encourage a return to private market funding sources.

In initially lending to AIG in September 2008 following the failure of Lehman Brothers, the Fed designed the Revolving Credit Facility to charge a premium even to the interest rate that a private sector consortium had proposed for itself just before the Lehman failure. In charging LIBOR plus 850 bps, the Fed hoped to limit moral hazard and encourage rapid repayment. However, AIG’s position continued to deteriorate, and the high interest rate (among other terms) portended further credit rating downgrades for AIG. Concomitant with other restructuring efforts, including a capital injection, the Fed began lowering the interest rate in November 2008—first by 550 bps—to ease AIG’s ability to repay and to reflect the stronger balance sheet that resulted from the other policy measures.

9. Loan Duration: How did AHEL lenders set the term of the loan?

Designers of modern emergency liquidity programs often emphasize designing programs to be “self-liquidating.”³⁸ Policymakers can then rely on this declining usage to determine the appropriate duration of a facility. That is, they suggest applying punitive terms relative to non-crisis-time market norms so that, when the acute phase of the crisis passes, borrowers are incentivized to return to private sources of liquidity. This approach certainly has merit in the cases of broad-based emergency liquidity provision, where the goal is to prevent or limit contagion from a marketwide event. Additionally, as such broad-based programs are designed to be marketwide, it’s not reasonable for the central bank to tailor the process of exiting the liquidity program to each specific borrower. However, this approach does not apply to cases of AHEL support—particularly given that it may not be appropriate to price AHEL support in a self-liquidating way (see Key Design Decisions No. 8, Rates and Fees, and No. 10, Balance Sheet Protection). Moreover, a firm in need of an ad hoc solution may see its problems outlive a crisis’s acute phase; removing the firm’s AHEL support prematurely may thus reignite market concerns.

An AHEL program’s duration should instead be tied to borrower-specific outcomes. Typically, this means tying duration to other policy outcomes. (As discussed in Key Design Decision No. 2, Part of Package, AHEL programs are almost exclusively bridges to, or support for, other policy measures rather than cures in themselves.) For instance, the Bank of Spain

³⁸ See, for example, Hauser (2021), Johnson and Santor (2013), Sack (2010), and Schulhofer-Wohl (2020).

left its AHEL line to CCM in place, supported by an ongoing fiscal guarantee, until CCM's restructuring plan could be implemented. At this point, the "good bank" of a good bank–bad bank restructuring took on the loan and paid it off. The Bank of England AHEL interventions for HBOS and RBS were left in place until both institutions could access government recapitalization and, for HBOS, until it could merge with Lloyds TSB.

However, the intent to tie duration to these outcomes should also be communicated to the extent possible—to help reduce market uncertainty. For Citi and BofA, which received contingent AHEL support during the GFC as a backstop to other fiscal guarantees, the duration was ultimately based upon the duration of the banks' problem assets that prompted the AHEL programs. The fiscal guarantees were based on the type of assets—10 years for residential mortgage assets, which were under particular market pressure in the GFC, and five years for all other assets—and the AHEL duration mirrored that. Thus, in the cases of Citi and BofA, the AHEL program duration was tied to the other policy measures, which were themselves based on the specific economic interactions of the current crisis and these banks' balance sheets. Policymakers also explicitly communicated these duration terms, so the market didn't have to worry about expiry approaching before the borrowers had steadied.

In the AIG RCF case, policymakers initially failed to establish an appropriate duration to allow for AIG to proceed with its plans to sell various businesses in an orderly way. They needed to extend the original two-year term to five years, which would be more conducive to both market stability and AIG's ability to repay the Fed.

When providing AHEL assistance to Bear Stearns, the policymakers did not initially clarify for Bear that the Fed's liquidity assistance on Friday was contingent upon Bear's selling itself over the weekend. Bear initially proceeded as though it had 28-day liquidity—and thus considerable breathing room—consistent with the Fed Board's authorization for the AHEL assistance to last up to 28 days. However, the Fed and Treasury were intent on the loan's being just a bridge to a weekend sale of Bear to JPM—owing to policymakers' view that Bear needed a guarantor of its obligations by Monday.

Policymakers may feel obligated to avoid tying AHEL duration to specific outcomes or deadlines if the possible outcomes—or the market's concerns over the institution—are not yet well understood at the time of the AHEL intervention. In such a case, policymakers should still provide assurance that the AHEL program will be in place as long as necessary to stabilize the institution by other measures.

For instance, the DNB did not establish or announce a fixed maturity on the AHEL assistance to Roskilde. While the DNB was determined to find a solution for Roskilde within six months, the DNB said only that it had "decided to provide the necessary liquidity" (DNB 2008). The Sveriges Riksbank said that its assistance to Carnegie was available "until further notice" (Riksbank 2008a; Riksbank 2008b).

At the opposite end of the spectrum, Bank Indonesia's regulations dictated that liquidity support for Bank Century was renewable only up to 90 days. By the time 90 days had passed, the government deposit insurer had seized the bank, and the deposit insurer was forced to

repay the loan to Bank Indonesia instead of the loan's being able to remain in place. Optimally, an AHEL program would be replaced with resolution financing when a bank is put into resolution; terms appropriate for an AHEL program may not be appropriate for longer-term resolution financing, particularly if the central bank is now facing another arm of government, as in the Indonesia case.³⁹

10. Balance Sheet Protection: How did the AHEL lender mitigate risks to its own balance sheet from provision of liquidity assistance?

As discussed in the Introduction and throughout this survey, AHEL programs' primary function is to provide the borrowing institution with a liquidity bridge until structural fixes to its nonviability can be made. As those additional measures are policy choices, or at least impacted by such choices, the institution's viability is at least partially endogenous to the crisis-fighters' other interventions.

These other policy measures and their intended outcomes can thus provide security to an AHEL lender beyond its ability to obtain "good collateral." This contrasts with broad-based emergency liquidity programs, where the soundness of collateral often serves as a proxy for the soundness of the borrower. Collateral is certainly the "easiest" way for the AHEL lender to protect its balance sheet, as it can relieve the lender of more uncertain assessments around the borrower's viability or the effects of other policy measures. Moreover, as discussed in Key Design Decision No. 1, Purpose, an AHEL program is often implemented when a borrowing institution's remaining collateral isn't eligible to be posted at standing facilities; expanding collateral eligibility by rolling out an AHEL facility can help relieve market pressure.

However, relying on collateral alone may limit or slow the AHEL lender's ability to provide an otherwise appropriate amount of liquidity assistance. Additionally, a central bank's use of (over)collateralized lending pushes down the seniority of other creditors, such as depositors, and may thus increase the incentive for the remaining funding to run. As former Bank of England Governor King writes of this phenomenon, "In extreme cases, the [liquidity assistance] is the Judas kiss for banks forced to turn to the central bank for support" (King 2016, 205).

Collateralizing an AHEL Facility

Broadly speaking, official-sector lenders loosen their collateral eligibility requirements as much as necessary to provide sufficient AHEL funding to the borrowing institution. For instance, after expanding in October 2007 the ad hoc liquidity available to Northern Rock—before ultimately lending in an amount equivalent to about 25% of Northern Rock's balance sheet—Bank of England Governor King described the loan's collateral as all the assets of Northern Rock, "right down to the paper clips" (House of Commons 2008).

³⁹ See also Arda and Nolte (2025). For a discussion of this issue in the US context during the Banking Crisis of 2023, see Kelly (2024b).

When the Fed rushed to provide AIG with sufficient liquidity in September 2008, it primarily took as collateral the holding company's stakes in its operating subsidiaries, which the Fed viewed as having ongoing viability (see US-AIG RCF). Similarly, when lending to Carnegie Investment Bank in October 2008, the Riksbank took as its primary collateral all the holding company's shares in Carnegie, those Carnegie held in its subsidiaries, and those of an affiliate.

In all these instances, the central bank was essentially taking the entire business as collateral, as opposed to taking specific assets with market-based valuations and applying haircuts, as would be standard central banking practice.

When taking more specific financial assets as collateral, AHEL lenders may get the most positive market reception by specifically targeting asset classes about which the market has become uncertain. During the GFC—which largely centered on securitization markets, and particularly those for housing—the Bank of England opened up AHEL facilities for Northern Rock, RBS, and HBOS that were specifically targeted at funding the assets they had in their securitization pipelines, especially unsecuritized mortgages. Moreover, lending against the assets most disfavored by the market can free up more-favored collateral for private lending or otherwise discourage taking additional losses through fire sales of the disfavored assets. However, purposely targeting an asset class that the market has abandoned is not a decision to take lightly, and the AHEL lender should take care to be confident that it is truly stepping in as emergency liquidity provider in an oversold market, rather than propping up a failed asset class. As the Bank of England wrote in a consultative paper in October 2008,

although [the BoE] may lend against securities for which private markets have closed suddenly in order to reduce the wider economic costs of the banking system's necessary adjustment, the Bank would wish to avoid permanently underpinning the existence of markets that were not fundamentally viable. (BoE 2008)

The inherent challenges to this kind of lending are the central bank's lack of preexisting familiarity with the asset class and ability to value an asset class made particularly illiquid by crisis. Both can challenge a central bank's ability to provide substantial funds against these assets. Central banks most typically and effectively lend against assets with established market prices or for which they have preexisting valuation processes in place.⁴⁰

However, a central bank is not typically capital- or liquidity-constrained and can thus be a particularly patient asset holder—and is often itself attempting to put a floor under asset prices in situations where it's providing liquidity against collateral that it views as oversold

⁴⁰ As a BIS working group notes in a 2017 paper, many central banks, after expanding eligible collateral during the GFC, have retained permanently more expansive collateral schedules,

recognising that a core function of central banks is to provide liquidity to solvent financial intermediaries facing temporary liquidity stress, and that the financial risks associated with accepting lower-grade collateral can be mitigated via exposure limits and appropriate haircuts. . . .

If collateral requirements are disclosed to institutions that might be eligible to participate in the facilities for [liquidity assistance], operational and financial risks can be further mitigated by regularly testing the facilities, by pre-pledging assets with the central bank, or by allowing the central bank to evaluate such assets before [they are pledged as collateral]. (BIS CGFS 2017)

by fire sales. Thus, it can be partially collateralized by the built-in discounts provided by the assets' depressed market prices.⁴¹

The fiscal authority can simplify the borrowing institution's ability to access central bank liquidity by performing collateral swaps or funding a capital injection with securities that the institution can use as collateral to borrow at the central bank. For instance, Latvia's ministry of finance provided Parex with government debt securities as term deposits, allowing Parex to, in turn, use those securities as collateral to borrow from the central bank.⁴² However, this option may also risk an undesirable bank-sovereign nexus. For instance, although Cyprus provided Laiki Bank with a capital injection through a transfer of government bonds, Laiki soon couldn't use them to borrow from the ECB, as Cyprus's sovereign credit rating fell below investment grade. While Laiki then moved to the CBC's ELA, the ECB soon exerted its ability to cut off the CBC from providing its own ELA as the ECB had concerns about Laiki's solvency. When Anglo Irish approached exhaustion of even non-ECB eligible collateral in 2010, the Irish finance ministry recapitalized the firm with sovereign debt. While this enabled Anglo to continue borrowing from the Central Bank of Ireland, the additional debt accreted to Ireland's sovereign crisis.

Perfecting a collateral interest can help the central bank ensure its seniority in the borrower's liability stack and thus further secure itself. Seniority often functions as one of the AHEL lender's primary means of balance sheet protection. When an AHEL borrower fails, is restructured, or is sold, AHEL lenders typically retain sufficient seniority to be transferred to an acquirer (or "good bank" in a good bank–bad bank solution) or otherwise be protected and paid back (see, for example, Cyprus–Laiki, Ireland–Anglo Irish, and Spain–CCM).

However, in some AHEL interventions, the borrowing entity has free collateral balances but cannot post them to the central bank as quickly as it needs the funds. When the SNB provided AHEL support to Credit Suisse, for instance, much of Credit Suisse's available collateral was located in non-Swiss entities. Additionally, even if the collateral transfer could have happened in time to collateralize the AHEL loan, transferring excessive amounts of collateral between entities would have tripped other regulatory triggers. The SNB's solution was to simply take preferential rights in bankruptcy as security for its "ELA+" AHEL facility.⁴³ This decision allowed the SNB to lend with seniority over unsecured creditors without Credit Suisse's having to first transfer and post collateral (FDF 2023; Tucker 2024). While not a case study surveyed here, the Fed's discount window assistance to Signature Bank on Friday, March 10, 2023 was of a similar flavor. The collateral Signature wanted to post to the Fed was tied up at Signature's FHLB lender and could not be transferred expeditiously enough for the Fed to perfect its collateral interest and still make a timely loan. To make the loan, the

⁴¹ See, for example, Alvarez (2022), Geithner (2014), and GAO (2011b).

⁴² This decision also shifted the counterparty risk of the borrowing from the central bank to the fiscal authority, a decision that was later validated in part by Parex's insolvency and subsequent recapitalization and restructuring—much of the emergency liquidity was later converted into equity.

⁴³ The parallel public liquidity backstop facility also had priority in bankruptcy, but with the added protection of a fiscal guarantee.

Fed accepted the FHLB's simply subordinating its own claim on the collateral it held to the Fed (NYDFS 2023).

When lending to Kaupthing, the CBI took as collateral all shares of Kaupthing's Danish subsidiary, FIH Erhverbsbank A/S. However, the cross-border nature of the CBI's assumption of FIH resulted in regulatory difficulties, wherein the CBI was unable to obtain direct ownership of the collateral.

Protection Provided by Forthcoming Measures

As discussed in Key Design Decision No. 3, Legal Authority, AHEL lending may be restricted by legal requirements that a borrower be "solvent." Some may interpret solvency requirements as a point-in-time accounting measure, but this is likely not an appropriate measure of a borrower's viability as a going concern—nor is it practical to perform such an assessment during a fast-moving crisis (Dobler et al. 2016; Goodhart 1999). When lending to a firm that remains viable or that will be viable once it is reinforced by other policy measures, the AHEL lender need not implement as much of its protection through collateralization. In the cases surveyed here, AHEL lenders regularly incorporated the added protection from other factors when assessing their level of security in lending.

By urging and facilitating a rapid merger with the healthier UBS in 2023, Swiss officials reduced the risk associated with the SNB's large AHEL loans to Credit Suisse. As the independent experts' report commissioned by the Swiss finance ministry assessed, "Once the merger was complete, UBS was liable for the loans, which contributed to an additional reduction of the risk" (FDF 2023).

The Fed's decision to provide a bridge loan to the Lehman Brothers broker-dealer LBI benefited from a similar rationale. The Fed had previously deemed that lending to the Lehman Brothers parent was inappropriate due both to assessing it to be nonviable—particularly as it lacked a buyer—and to fears about its collateral, particularly given the likelihood the Fed would have been forced to manage it all. Nonetheless, the Fed was comfortable providing AHEL assistance to LBI in part because LBI was working toward a likely deal to be purchased by Barclays. That the loan was clearly functioning as a bridge loan offered the Fed additional protection, as the Fed would be facing Barclays after the acquisition (Bernanke 2015a; Geithner and Metrick 2018; Valukas 2010).

Returning to the RBS and HBOS cases, the Bank of England provided AHEL assistance based on its observation of a clear path to future solvency due to other policy measures. HBOS had announced a merger with a stronger institution, Lloyds TSB; for RBS, the BoE anticipated an impending capital injection from a fiscal recapitalization program that was, at the time of the AHEL provision, almost complete. In the HBOS case, the BoE had even contemplated lending on an unsecured basis if it became necessary as a bridge to the Lloyds merger and pursuant new funding plan (Plenderleith 2012).

The Central Bank of Cyprus AHEL assistance to Laiki Bank from 2011–2013 should offer a note of caution, however. In its ongoing solvency assessment of Laiki, the CBC relied upon a preliminary Memorandum of Understanding (MoU) between the Cyprus government and

international financial authorities. The MoU would have provided a financial assistance package to Cyprus inclusive of EUR 10 billion to recapitalize Cypriot banks. Yet, this MoU ultimately did not reach final agreement until after Laiki was in resolution.

Protection Provided by the Fiscal Authority

Beyond policies designed to ensure borrower viability, constraints owing to the borrower's solvency or collateral can in some cases be alleviated by fiscal guarantees (see Key Design Decision No. 3, Legal Authority, and Arnold [2025a]). In eight of the 22 underlying cases surveyed, an ad hoc state guarantee of the AHEL lending served as a complement to collateral or to provide additional lending capacity. For instance, the BoE obtained a state guarantee for some of its lending to Northern Rock, but only on amounts lent after a particular date. Half of the CHF 200 billion (USD 217 billion) of AHEL assistance that the SNB made available to Credit Suisse came with a government guarantee. After borrowing CHF 48 billion from standing SNB facilities and CHF 50 billion from the ELA+ facility, Credit Suisse borrowed its next CHF 70 billion from the SNB's PLB—an AHEL facility identical to ELA+ but with a fiscal guarantee for the SNB (both facilities were available for up to CHF 100 billion). More aggressively, the DNB required no collateral from Roskilde for its AHEL loan, which was guaranteed by the state (behind a small initial tranche guaranteed by a standing, banking industry rescue group).

Broadly speaking, fiscal guarantees of central bank lending may also provide the central bank with desirable political insulation on particularly visible or large loans, especially those with greater potential fiscal implications.⁴⁴ For instance, even when fiscal resources were not available to protect the Fed in its AHEL assistance to Bear and AIG, the Fed requested the Treasury secretary write public letters of support. The letters specifically acknowledged that any losses on these AHEL operations would reduce Fed earnings remittances to the Treasury—thus having fiscal implications (Paulson 2008a; Paulson 2008b; Paulson 2010, 114–15). These letters simply stated fiscal facts and fell short of Treasury indemnity, but the Fed viewed them as still providing some degree of political cover from the Treasury (Geithner 2014, 156).⁴⁵

11. Impact on Monetary Policy Transmission: How did central banks manage the impact of the AHEL lending on their monetary policy stance?

When it comes to an AHEL program's impact on monetary policy transmission, a central bank's usual calculus when injecting central bank reserves applies. Any monetary policy impact depends on both the relative size of the intervention and the relative size of the affected institution to the size of the economy. As discussed in Key Design Decision No. 1, Purpose, AHEL programs are typically executed owing to perceptions of systemic risk in their absence; to the extent a financial institution's systemic risk is already correlated with

⁴⁴ For a detailed discussion of fiscal protection of central bank emergency lending in the US during recent crises, see Kelly (2025).

⁴⁵ Notably, the Treasury secretary also wrote such a letter to the Fed in the US–Citigroup case, even though the Fed loan facility was explicitly junior to then-available fiscally allocated funds (Paulson 2009).

size, central bankers should keep an eye toward the potential need for sterilization operations.

After the Fed implemented its Bear Stearns rescue in 2008, which was accompanied by several other expansionary monetary operations, the Fed—to prevent its monetary policy interest rate from falling below its target rate—began sterilizing its liquidity operations through sales of Treasury bills from its monetary policy portfolio. By the time of its AHEL assistance for the Lehman broker-dealer (LBI), the Fed had to begin involving the Treasury in its sterilization efforts because it could sterilize only up to the size of its preexisting monetary policy portfolio. In the Supplementary Financing Program, the Treasury issued short-term bills and left the proceeds at the Fed (Fed 2009).

The monetary impacts of unsterilized lending can in some instances be significant. In the Moldova–Consortium case, the NBM’s rapid expansion of reserves by more than 50% contributed to a near doubling in inflation. The NBM itself said that its AHEL assistance represented a “monetary issue,” as prices rose and the Moldovan foreign exchange rate plummeted (NBM 2020). Beyond such an extreme case, sterilization may be important from a monetary policy communications perspective. Adding reserves via an AHEL intervention may be interpreted by the market as a signal of an easing monetary stance, which is undesirable when such a signal is at cross-purposes with the central bank’s monetary policy goals; for an example in the case of a BBEL and market support operation (the Bank of England’s gilt market intervention in 2022), see Hinge (2022).

Rather than outright sales of portfolio assets, central banks may also structure liquidity assistance as an asset swap. Collateral swaps, by avoiding adding new reserves, may help limit the impact on monetary policy transmission, but, as with asset sales sterilization efforts, they are fundamentally limited by the size of the central bank’s preexisting asset portfolio.

Some central banks also have the authority to issue their own bills, which can offset any unintended monetary easing. When the SNB opened two AHEL facilities for Credit Suisse in 2023, their combined caps exceeded 25% of Swiss GDP.⁴⁶ The SNB used both the issuance of SNB bills and reverse repo operations to partially offset the AHEL injections and prevent the assistance from altering the SNB’s monetary stance. Sweden’s Riksbank similarly tightened interbank liquidity following its 2008 AHEL provision via a combination of normal monetary policy operations and the issuance of one-week Riksbank bills.

It should also be noted that the increasing dearth of so-called scarce reserves monetary policy regimes since the GFC may obviate much of the need to worry about sterilization of AHEL interventions. In regimes where reserves are ample and policy rates are simply administered, reserve creation is automatically sterilized by the interest paid on reserves.

Similarly, AHEL assistance provided by non-central bank lenders does not increase the supply of reserves; depending on how it is funded, though, it may still impact interest rate markets. Lastly, it is worth noting that to the extent that interbank lending has broadly

⁴⁶ This value does not include the additional liquidity drawn from the SNB’s standing facilities.

retreated due to a crisis, the quantum of reserve creation is likely less predictive of the impact on monetary policy transmission—that is, a given quantity of reserve creation will likely have less inflationary impact than it would in a noncrisis environment.

12. Other Conditions: What additional conditions did the AHEL lender attach to the assistance?

As noted in Key Design Decision No. 2, Part of Package, AHEL assistance is almost always a bridge to—or a simultaneous part of—more structural policy responses. The receiving banks are often mandated to do things such as remove management, refrain from making shareholder distributions, or submit to heightened surveillance. Yet, it is difficult in many cases to disentangle whether such conditions are due to the AHEL assistance or those other policy measures, such as restructurings or capital injections.

As discussed throughout this survey, when implementing punitive conditions for ad hoc assistance, there is a trade-off between moral hazard and the institution's viability (or at least perceptions thereof). If a borrowing institution accepts a particularly punitive AHEL loan, its willingness to do so may unproductively raise the market's perception of the severity of the firm's problems. Punitive conditions may also encourage the exodus of employees and clients. While, say, hasty firings or compensation cuts may reduce costs and managerial incompetence, they may also lead to erosion of franchise value and to further market pressure on the institution. For instance, after the bank run on Northern Rock, each of its board members signed an agreement expressing willingness to resign if needed. However, these agreements were not utilized at the time due to feedback from the bank's stakeholders that they wanted the current board to first manage the existing crisis (House of Commons 2008). Thus, punitive additional conditions should be carefully considered and likely implemented only in the context of crisis-fighters' having other measures in place to ensure the institution's ongoing viability.

Relatedly, policymakers should be conscious of the risk of these other loan conditions discouraging potential acquirers. Being acquired by a healthier institution can be a relatively quick solution for a bank under pressure; the Bear Stearns, Roskilde, and AIG RCF cases of AHEL assistance were conditioned on the borrowing bank's finding a buyer for all or parts of itself. Originally, the conditions of the assistance for Credit Suisse included a provision that a borrower under the program could not make shareholder distributions; however, officials amended this provision on the day of the announced merger with UBS to allow for an exception if the borrower was taken over by another entity (SFC 2023a; SFC 2023b).

Conclusion

This paper surveys 22 modern cases of ad hoc emergency liquidity intervention. AHEL interventions are a useful tool for policymakers to meet the liquidity demands of an institution facing a run of systemic consequence in relatively quick order. Yet, as preeminent central banker and former Fed chair Paul Volcker has said, "I long ago came to the conclusion there are very few liquidity crises that aren't related to a solvency question. There are few

occasions where people aren't willing to lend unless they have a suspicion about the solvency of the borrower" (Volcker 2008). Mervyn King, who helmed the Bank of England during the GFC, would later write similarly, "Almost every financial crisis starts with the belief that the provision of more liquidity is the answer, only for time to reveal that beneath the surface are genuine problems of solvency" (King 2016, 367).

As shown in the 21st century case studies surveyed here, this stylized fact continues to ring true—and could perhaps be made more accurate only by replacing the word "solvency" with the word "viability." Despite the ad hoc provision of liquidity, which in several instances was sized to meet all potential outflows, in no cases did liquidity provision alone prove a "cure" to the run on the institution. Instead, AHEL assistance allowed the borrower to continue to meet its obligations while it underwent a broader restructuring of its balance sheet.

Establishing an AHEL program with the hope that the institution can avoid insolvency as long as it is saved from illiquidity will almost certainly lead to wasted time or lost credibility for crisis-fighters. An effective AHEL program is one that is followed as expeditiously as possible by additional policy measures to address the borrower's chronic problems and restore it to viability—most commonly via a government capital injection or merger with a stronger institution.

Perhaps counterintuitively, greater up-front recognition from crisis-fighters that AHEL assistance will be followed by additional policy measures can enable them to provide the liquidity more effectively. Given that additional policy measures are to follow, the terms of an AHEL program can focus on providing sufficient funding and be less bound by concerns of moral hazard. Such terms as penalty interest rates and steep collateral demands do not make for a credible AHEL program if they accelerate the firm's financial deterioration or limit the firm's access to available funding. Moreover, ignoring pending changes to a borrower's balance sheet—such as a capital injection or a takeover by a healthier firm—makes little economic sense when determining borrowing terms.

The received wisdom on Bagehot's dictum assumes that the restrictions of good collateral and penalty rates are designed so that the central bank functions as a backstop to systemic liquidity demands.⁴⁷ AHEL programs are not backstops like their discount window and BBEL brethren; they must provide the funds to meet the run on the systemic institution. Interventions that trail the AHEL assistance to solve the borrower's chronic balance sheet problem can include terms intended to address moral hazard concerns.

⁴⁷ Consistent with the type of problem Bagehot was diagnosing, the dictum does offer at least a starting place for alleviating contagion through BBEL or market liquidity programs (Rhee, Feldberg, et al. 2020; Rhee, Feldberg, et al. 2022; Wiggins, Fulmer, et al. 2023). For instance, Wiggins, Fulmer, et al. 2023 restates the dictum for BBEL programs as "In the acute (panic) phase of a crisis, a [lender of last resort] should lend freely and broadly against good collateral, at rates set by auction, taking care to avoid disclosing details about individual borrowers."

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